

CS471 – Lesson 19 In-Class Worksheet

Probability: Bayes' Rule and Base Rate Reasoning

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Apply Bayes' Rule to update beliefs, compute intermediate joint and marginal probabilities, and evaluate the impact of base rates on posterior confidence.

1. Probability as Uncertainty and Joint Probabilities

- a. A Space Force satellite subsystem component has a probability of 0.80 of remaining operational through the next mission segment. In one sentence, interpret this probability in an operational context.

- b. For a battlefield reconnaissance system, adequate visibility occurs with probability $P(V) = 0.75$, and reliable communication given adequate visibility occurs with probability $P(C | V) = 0.80$. Compute the joint probability $P(V, C)$.

2. Bayes' Rule: Updating Beliefs Under Sensor Evidence

An autonomous satellite diagnostics system evaluates a critical propulsion valve failure (F).

- Prior probability of failure: $P(F) = 0.10$
 - Detector likelihood given failure: $P(A | F) = 0.80$
 - Marginal probability of an alert: $P(A) = 0.20$
- a. Write the Bayes' Rule formula used to compute the posterior probability $P(F | A)$, identifying the role of the prior, likelihood, and marginal evidence.

- b. Calculate the numerical posterior probability $P(F | A)$ when the detector triggers an alert.

3. Base Rate Reasoning and Independence

- An automated early-warning detector identifies 95% of actual incoming missile threats ($P(\text{Alert} \mid \text{Threat}) = 0.95$), but threats occur in only 1% of all operational radar sweeps ($P(\text{Threat}) = 0.01$). Explain why an alert might still yield a posterior probability far below 95%.
- Suppose diagnostic testing reveals $P(\text{Radar Fail}) = 0.30$ and $P(\text{Radar Fail} \mid \text{SATCOM Fail}) = 0.30$. Are these failure events independent or dependent? Justify your answer.

Part 2: Student Independent Practice

Objective: Formulate full Bayesian probability pipelines—calculating prior, likelihood, joint, marginal, and posterior values for tactical mission assurance.

A. Anti-Submarine Warfare (ASW) Sonobuoy Classification

An autonomous maritime patrol aircraft drops a sonobuoy field to detect adversary submarines in a contested chokepoint.

- **Prior Base Rate:** Hostile submarines are present in the sector during 4% of missions: $P(S) = 0.04$.
- **Sensor Sensitivity (Likelihood):** If a submarine is present, the sonobuoy triggers an acoustic alert (A) 90% of the time: $P(A | S) = 0.90$.
- **False Alarm Rate:** If no submarine is present ($\neg S$), ambient marine wildlife triggers a false alert 5% of the time: $P(A | \neg S) = 0.05$.

1. Joint Probabilities:

- a. Compute the joint probability of a true submarine being present and an alert triggering, $P(S, A)$.
- b. Compute the joint probability of clear water ($\neg S$) and a false alarm triggering, $P(\neg S, A)$.

2. Marginal Evidence Probability:

Using the Law of Total Probability, compute the total marginal probability that the sonobuoy triggers an alert, $P(A)$.

3. Posterior Probability (Bayes' Rule):

If the sonobuoy detects a contact and sounds an alarm (A), compute the updated posterior probability that a hostile submarine is actually present, $P(S | A)$.

B. Cyber Intrusion Detection System (IDS)

A military base network Intrusion Detection System (IDS) flags suspicious network traffic:

- Baseline rate of active Advanced Persistent Threat (APT) attacks: $P(\text{APT}) = 0.002$ (0.2%).
- True positive detection rate: $P(\text{Alarm} \mid \text{APT}) = 0.98$.
- False alarm rate on benign traffic: $P(\text{Alarm} \mid \neg \text{APT}) = 0.01$.

1. Compute the total marginal probability of an IDS alarm, $P(\text{Alarm})$.
2. Compute the posterior probability that an active APT attack is occurring given an alarm, $P(\text{APT} \mid \text{Alarm})$.
3. What operational conclusion should the Cyber Defense Watch Officer draw regarding whether to immediately disconnect the server based on an alarm?